

Translated

From: e-maxx.ru

The Stern-Brocot tree and Farey sequences

Stern-Brocot tree

The Stern-Brocot tree is an elegant construction to represent the set of all positive fractions. It was independently discovered by German mathematician Moritz Stern in 1858 and by French watchmaker Achille Brocot in 1861. However, some sources attribute the discovery to ancient Greek mathematician Eratosthenes.

The construction starts at the zeroth iteration with the two fractions

$$\frac{0}{1}, \frac{1}{0}$$

where it should be noted that the second quantity is not strictly a fraction, but it can be interpreted as an irreducible fraction representing infinity.

At every subsequent iteration, consider all adjacent fractions $\frac{a}{b}$ and $\frac{c}{d}$ and insert their **mediant** $\frac{a+c}{b+d}$ between them.

The first few iterations look like this:

$$\begin{array}{c} \frac{0}{1}, \frac{1}{1}, \frac{1}{0} \\ \frac{0}{1}, \frac{1}{2}, \frac{1}{1} \\ \frac{1}{2}, \frac{1}{1}, \frac{2}{1}, \frac{1}{0} \\ \frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1} \\ \frac{1}{3}, \frac{2}{3}, \frac{3}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{0} \end{array}$$

Continuing this process to infinity this covers *all* positive fractions. Additionally, all fractions will be *unique* and *irreducible*. Finally, the fractions will also appear in ascending order.

Before proving these properties, let us actually show a visualization of the Stern-Brocot tree, rather than the list representation. Every fraction in the tree has two children. Each child is the mediant of the closest ancestor on the left and closest ancestor to the right.